

Mycobacterium fortuitum Necrotizing Cellulitis after Dog Bite in a Pediatric Patient: a case report

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Case Report

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Abstract

Background

Mycobacterium fortuitum (*M. fortuitum*), a rapidly growing nontuberculous mycobacterium (NTM), is an emerging pathogen in trauma-associated skin and soft tissue infections (SSTIs). Pediatric NTM infections are rare and often misdiagnosed due to atypical presentations and intrinsic β -lactam resistance. We report the first pediatric case of *M. fortuitum* necrotizing cellulitis following a dog bite in an 8-year-old child, highlighting diagnostic and management challenges.

Case presentation

: An 8-year-old child presented with necrotizing cellulitis following a dog bite, which was refractory to empiric β -lactam therapy. Histopathology revealed nodular granulomatous inflammation with epithelioid cells and purulent necrosis, and acid-fast bacilli were identified via Ziehl-Neelsen staining. Metagenomic next-generation sequencing (mNGS) rapidly identified *M. fortuitum* within 48 hours, confirmed by automated liquid culture (7-day growth) and targeted PCR. A multidisciplinary approach integrating broth microdilution-guided combination therapy (amikacin 0.5 g three times weekly, linezolid 0.3 g twice daily, and doxycycline 0.1 g twice daily) with weekly or biweekly surgical debridement achieved complete resolution of cutaneous lesions.

Conclusions

This case highlights three key points: (1) NTM infections should be considered in animal bite wounds unresponsive to β -lactams; (2) mNGS accelerates pathogen detection, enabling timely susceptibility-directed therapy; (3) Combined antimicrobial and surgical strategies are essential for successful treatment. Early multidisciplinary consultation is crucial to prevent chronic infection or systemic dissemination in complex SSTIs.

Clinical trial number:

Not applicable.

Background

Animal bites are a significant global health issue, accounting for 5% of trauma-related emergency visits, with dog bites representing 80–90% of cases [1, 2]. Pediatric populations are particularly affected, with 337,000 annual U.S. emergency visits and \$2 billion in healthcare costs [3]. While only 5–25% of bite injuries develop significant infections [4], complications such as rabies prophylaxis and antimicrobial-resistant skin and soft tissue infections (SSTIs) place a substantial strain on healthcare systems. The

microbiology of bite wounds involves a mix of canine oral commensals, human cutaneous microbiota, and environmental organisms, often leading to polymicrobial infections [5, 6]. Delayed presentations may involve opportunistic or fastidious pathogens, including NTM, which are challenging to diagnose and treat. *M. fortuitum*, a rapidly growing NTM with intrinsic resistance to β -lactams and macrolides, exemplifies this challenge. Its clinical presentation mimics common pyogenic infections, and cultures yield positive results in only 40%-60% of confirmed cases [7]. Effective management requires prolonged, susceptibility-guided combination therapy [8]. This study reports the first pediatric case of *M. fortuitum*-induced necrotizing cellulitis following a dog bite, highlighting diagnostic pitfalls and therapeutic strategies. It underscores the need for early multidisciplinary consultation in managing complex SSTIs. We present this case in accordance with the CARE reporting checklist.

Case Presentation

An 8-year-old girl (height 140 cm, weight 34 kg) was admitted on December 12, 2023, with a non-healing right thigh wound complicated by sinus tract formation and continuous exudate, persisting for 2 months after a domestic dog bite. Initial treatment at an external hospital included tetanus immunoglobulin, rabies vaccination, and wound irrigation with suturing, but the wound failed to heal. Imaging (MRI/ultrasound) revealed a subcutaneous abscess (3×2 cm) with surrounding soft tissue edema and ipsilateral inguinal lymphadenopathy (short-axis diameter 1.0 cm). Wound cultures showed no pathogenic bacterial growth.

On admission, physical examination revealed two residual dog bite wounds with purulent discharge from a central sinus tract, erythema (4×3 cm), and tenderness in the right inguinal region. Laboratory tests showed white blood cell count (WBC) $7.8 \times 10^9/L$ and neutrophil percentage (NEUT%) 65%. On December 13, she underwent debridement, inguinal lymphadenectomy, and vacuum-sealed drainage (VSD). Post-operatively, she developed intermittent fever and was treated with intravenous amoxicillin-clavulanate potassium (1 g every 8 hours). The clinical course is shown in Fig. 1. Histopathology examination of the surgical specimens revealed epithelioid cell granulomatous inflammation with purulent necrosis (Fig. 2). Acid-fast staining was positive (3–9 bacilli per 100 fields), but the tuberculosis antibody test was negative. One week later, during the second debridement, the intraoperative specimen tested negative for *Mycobacterium tuberculosis* DNA.

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On December 27, during the third debridement, metagenomic next-generation sequencing (mNGS) identified *Mycobacterium fortuitum* within 48 hours (sequence count 14,155; confidence level 99%) (Fig. 3). Following consultation with a clinical pharmacist, a treatment regimen was initiated, including oral clarithromycin (0.5 g once daily), linezolid (0.3 g twice daily), and intravenous amikacin (0.5 g three times weekly). However, wound healing was not as expected. Two weeks later, tissue from re-debridement was cultured for 7 days, and *M. fortuitum* was confirmed by automated liquid culture and PCR, with antimicrobial susceptibility results obtained (Table 1). The clinical pharmacist recommended replacing clarithromycin with doxycycline. On January 17, during the fifth debridement, necrotic tissue

decreased but acid-fast staining remained positive. Cefmetazole (1 g twice daily) was added, and doxycycline was discontinued after 21 days due to elevated liver enzymes (alanine aminotransferase [ALT] 80.0 U/L and aspartate aminotransferase [AST] 55.9 U/L). Two weeks later, during the sixth debridement, the wound showed abundant granulation tissue, and liver enzymes had normalized.

The patient was discharged on February 12 and continued oral linezolid for 3 months. Follow-up showed good wound healing with no discomfort.

This study was approved by the Ethics Review Committee of the First College of Clinical Medical Science, China Three Gorges University (Ethics No. 2024-251-01). All procedures performed in this study were in accordance with the ethical standards of the institutional and/or national research committee and with the Helsinki Declaration (as revised in 2013). Written informed consent was obtained from the patient's parents for publication of this case report and accompanying images. A copy of the written consent is available for review by the editorial office of this journal.

Discussion

This case provides the first molecular evidence confirming a direct causal link between dog bites and *M. fortuitum* infection, offering critical clinical evidence for the zoonotic transmission mechanism of NTM. As environmental commensals, NTM are primarily transmitted through soil or water exposure, with previous consensus suggesting no animal-to-human transmission [9]. Although domestic animals can harbor NTM and develop self-infections [10, 11], human infections resulting from animal bites are exceedingly rare, with only five documented cases of canine-derived infections in the literature [12–16]. This case, supported by existing evidence, demonstrates that oral NTM colonization in canines constitutes a novel transmission risk. The underlying mechanisms involve traumatic inoculation leading to biofilm formation, dysbiosis of the salivary microbiome promoting proliferation of opportunistic pathogens, and immunodeficiency-related opportunistic infections [5, 6]. Dogs may act as potential zoonotic vectors, underscoring the necessity for clinical reevaluation of infection risks associated with animal-related trauma.

The clinical manifestations of NTM skin infections are relatively insidious, posing significant diagnostic challenges. With an incubation period of 2–4 weeks, the pathological presentation involves granulomatous inflammation combined with suppurative changes, which overlaps histologically with tuberculosis infection. Traditional diagnostic methods for NTM—such as acid-fast staining (with low sensitivity), prolonged culture cycles (4–6 weeks), and phenotypic differentiation difficulties [8, 17]—highlight the limitations of conventional diagnostic systems. In this case, mNGS was employed to accurately identify the pathogen within 48 hours. As a hypothesis-free and unbiased approach, mNGS leverages high-throughput and rapid sequencing capabilities to detect and quantify all microorganisms in a single specimen. Quantitative analysis effectively distinguishes pathogens from colonizing flora, as demonstrated here by the detection of 14,155 specific sequences of *M. fortuitum*, significantly surpassing background microbial signals. This technology overcomes the "blind spots" of clinical

presumption, shortens the identification time for rare pathogens, and provides critical evidence for early targeted therapeutic decisions [18, 19]. We recommend incorporating mNGS as an alternative diagnostic tool for non-healing wounds (> 7 days) following animal bites or cases refractory to conventional treatments.

The treatment of NTM skin infections requires individualized regimens integrating antimicrobial susceptibility testing, tissue penetration profiles, and patient-specific factors [10]. *M. fortuitum* generally exhibits resistance to first-line antituberculosis drugs, with amikacin demonstrating the strongest in vitro antibacterial activity, while susceptibility varies for ciprofloxacin, linezolid, imipenem, doxycycline, and clarithromycin [20–22]. For pediatric patients, priority is given to oral agents with high bioavailability and effective cutaneous tissue concentrations, while balancing safety considerations: avoiding fluoroquinolones due to skeletal toxicity risks and restricting carbapenems to maintain intestinal microbiota equilibrium. In this case, empirical therapy with clarithromycin, linezolid, and amikacin was initially administered. The regimen was subsequently optimized based on susceptibility results, replacing clarithromycin with doxycycline due to macrolide resistance mediated by *erm(39)* gene-induced methylation of the 23S rRNA macrolide-binding site [23]. A pediatric-specific risk control system was implemented, including low-dose intermittent amikacin administration [24, 25] combined with otoacoustic emission monitoring to mitigate cumulative dose-related ototoxicity and nephrotoxicity, alongside doxycycline therapy limited to ≤ 21 days [26]. The dynamic therapeutic strategy featured an intensive intravenous combination phase lasting 6 weeks, followed by an oral step-down consolidation phase for 12 weeks [8, 27]. Additionally, this strategy is supported by six surgical debridements: weekly during the acute phase and biweekly during the chronic phase. Therapeutic efficacy was determined through dual quantitative criteria: three consecutive negative wound cultures and $\geq 80\%$ granulation tissue coverage, ultimately achieving complete wound healing through this four-month multimodal intervention.

Conclusions

This case underscores the risk of NTM infections from dog bites and highlights mNGS's role in rapid diagnosis. Effective treatment involves susceptibility-guided antimicrobial therapy and surgery, with vigilant monitoring for adverse effects. Given the limitations of current guidelines and this single-case report, further multicentric studies are needed to refine treatment protocols. Clinicians should be alert to zoonotic NTM transmission and consider it in refractory cases.

Declarations

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Author contributions

(I) Conception and design: Z Mei, M Xia; (II) Administrative support: G Jin; (III) Provision of study materials or patients: Z Mei, M Xia; (IV) Collection and assembly of data: Y Tan, T Zheng; (V) Data analysis and interpretation: Z Mei, M Xia; (VI) Manuscript writing: All authors; (VII) Final approval of manuscript: All authors.

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Data availability

No datasets were generated or analysed during the current study.

Ethical approval and consent to participate

This study was approved by the Ethics Review Committee of the First College of Clinical Medical Science, China Three Gorges University (Ethics NO. 2024-251-01). All procedures performed in this study were in accordance with the ethical standards of the institutional and/or national research committee and with the Helsinki Declaration (as revised in 2013).

Consent for publication

Written informed consent was obtained from the patient's parents for publication of this case report and accompanying images. A copy of the written consent is available for review by the editorial office of this journal.

Competing interests

The authors declare no competing interests.

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Table 1

Table 1. Antimicrobial susceptibility profile of *M. fortuitum* isolate according to CLSI M24 guidelines

Abbreviations: MIC, Minimum inhibitory concentration; S, Susceptible; I, Intermediary; R, Resistant.

Antimicrobial Drug	MIC, µg/mL	Interpretation
Amikacin	4	S
Rifampicin	>4	R
Rifambudin	4	R
Clarithromycin	16	R
Imipenem	8	I
Minocycline	1	S
Tigacycline	0.5	S
Trimethoprim	4/76	R
Amoxicillin	>64/32	R
Cefoxitin	16	S
Ciprofloxacin	≤0.125	S
Doxycycline	0.5	S
Linezolid	5	S
Moxifloxacin	≤0.25	S
Tobramycin	>16	R
Meropenem	4	S

Figures

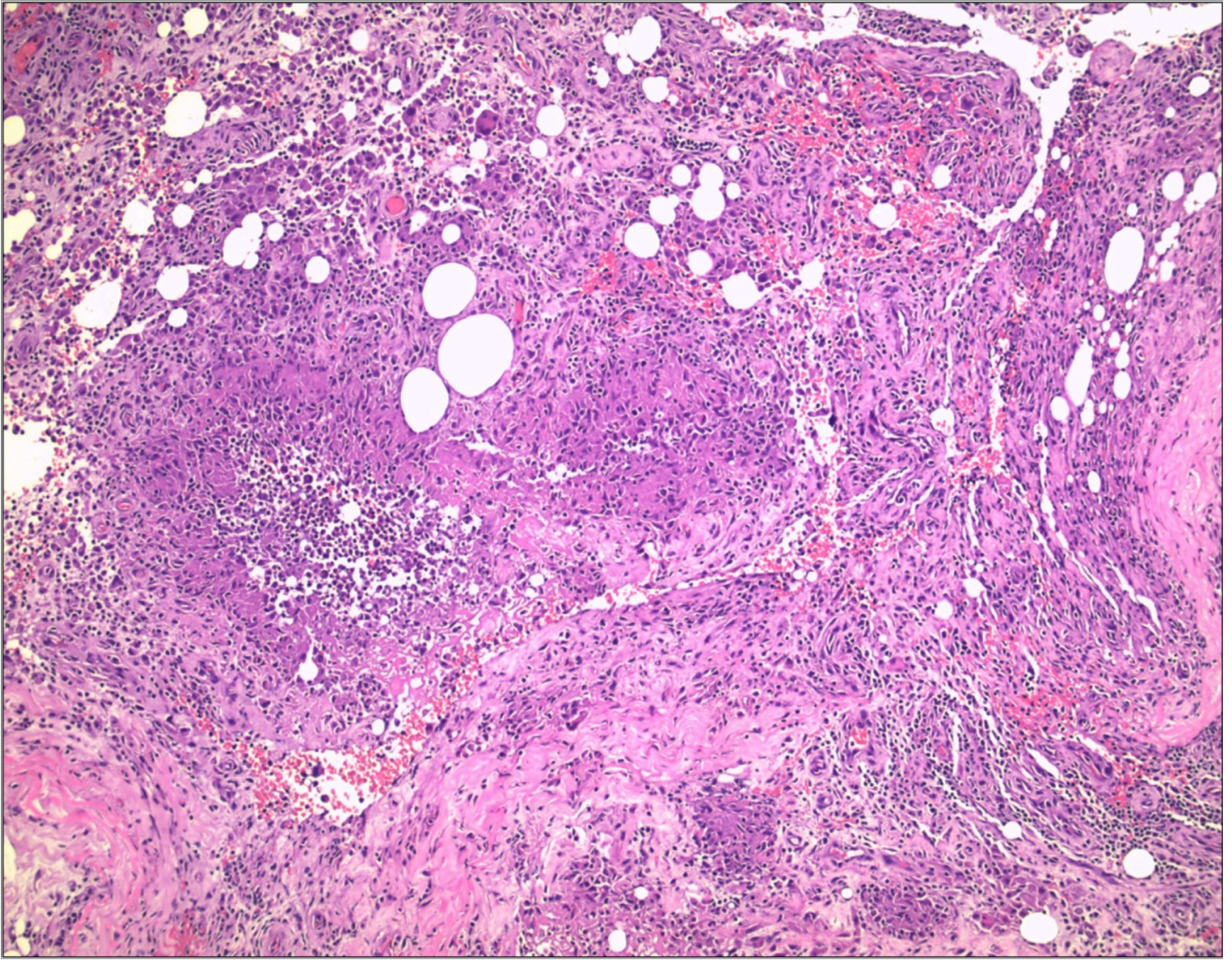


Figure 2

Histopathological characteristics of necrotic tissue. (Hematoxylin and eosin; x100.) Inflammatory pyogenic necrosis surrounded by nodular granulomatous epithelioid cells was observed.

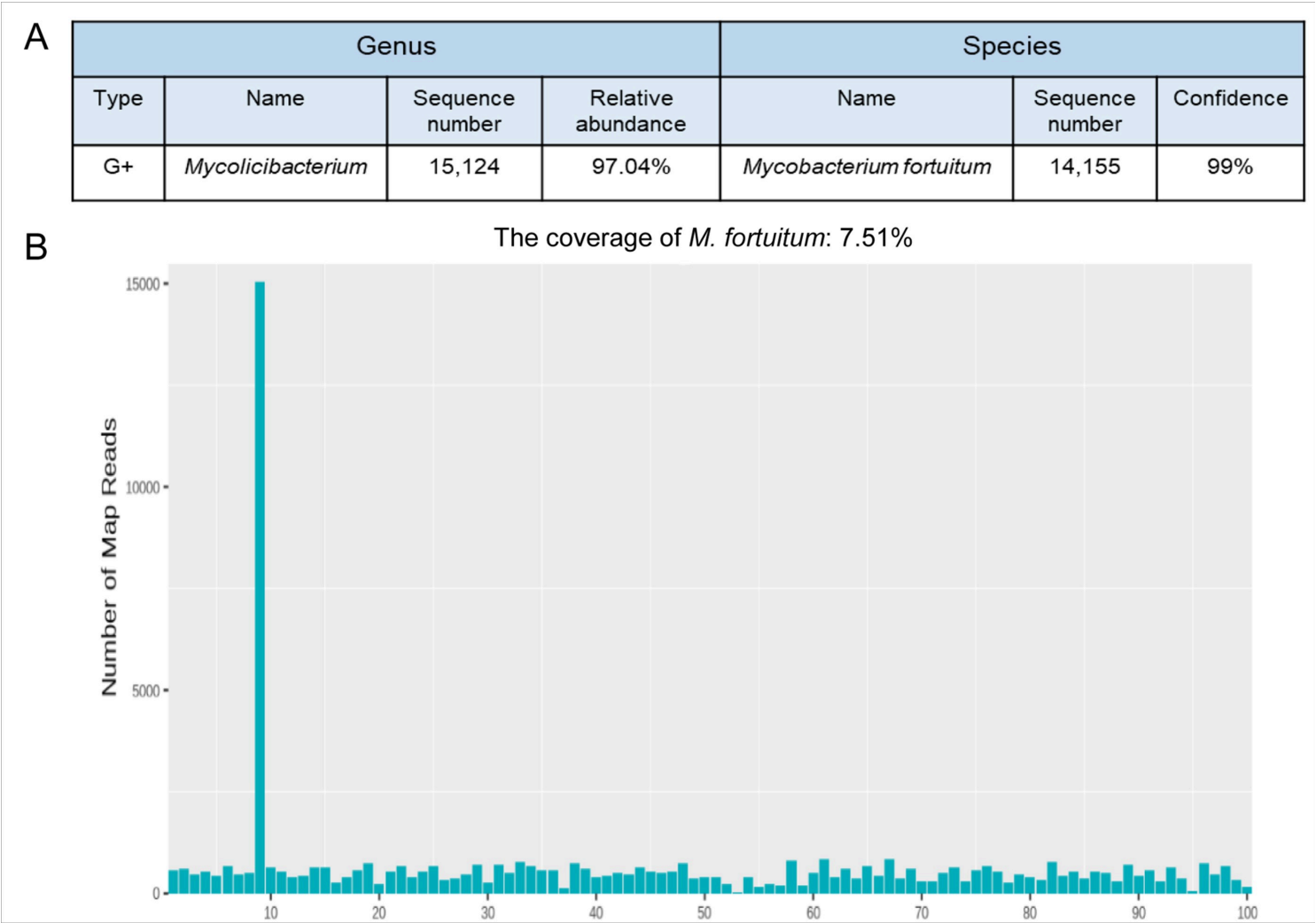


Figure 3

Genome coverage map of detected sequences of *M. fortuitum* by mNGS in necrotic tissues. (The coverage of *M. fortuitum*: 7.51%.)